Moumen Nermine
nermine.moumen@insat.ucar.tn | +216 27 040 414

Subject: Application for the CO2-Active Fischer-Tropsch Process Modeling Internship — Institute for Future Fuels

Dear Hiring Team,

I am a final-year Industrial Chemical Engineering student at INSAT (Université de Carthage, Tunisia), specializing in Process and Environment, and I am writing to apply for the internship on techno-economic assessment of the CO2-active Fischer-Tropsch route within the Solar Production Process Assessment department. I am available for a six-month internship starting February 1, 2027 (flexible between 3 and 9 months).

My coursework at INSAT has given me a strong grounding in renewable energy, environmental science, resource valorization, and sustainability, backed by the mathematics and thermodynamics needed to actually quantify them, a foundation I've also put into practice outside the classroom: with Enactus INSAT, I led PhosphAct, a project valorizing phosphogypsum waste into paint. This same techno-economic and environmental lens is what I've applied in my project work: a comparative Aspen Plus study of two ethylene production routes (steam cracking vs. bioethanol dehydration), combining reactor and separation-train design with a cradle-to-gate Life Cycle Assessment in OpenLCA and a cost evaluation via the Aspen Process Economic Analyzer, an evaluation workflow directly analogous to the reference-case benchmarking and techno-economic assessment this role requires. In a separate project, I modeled a PEM electrolyzer in Ansys Fluent, which gives me a working grasp of the electrolysis stage upstream of your process.

DLR's work on intensifying RWGS and Fischer-Tropsch onto a single catalyst is exactly the kind of process-intensification question I want to build my final-year project around. I don't yet have hands-on experience with Fischer-Tropsch catalysis specifically, but I have solved reaction-kinetics problems in Aspen (Langmuir-Hinshelwood kinetics for an ethylene oxide reactor) and numerically solved coupled PDEs for an electrochemical model during my current internship at SEGULA Technologies, calibrating parameters against experimental data which is the same kind of quantitative, self-directed modeling work this project calls for.

I am drawn to this internship because it sits precisely at the intersection I want to specialize in: renewable-energy-driven process simulation and its techno-economic viability.

I would welcome the opportunity to discuss my application further and am available for an interview at your convenience. Thank you for your consideration.

Sincerely,
Nermine Moumen